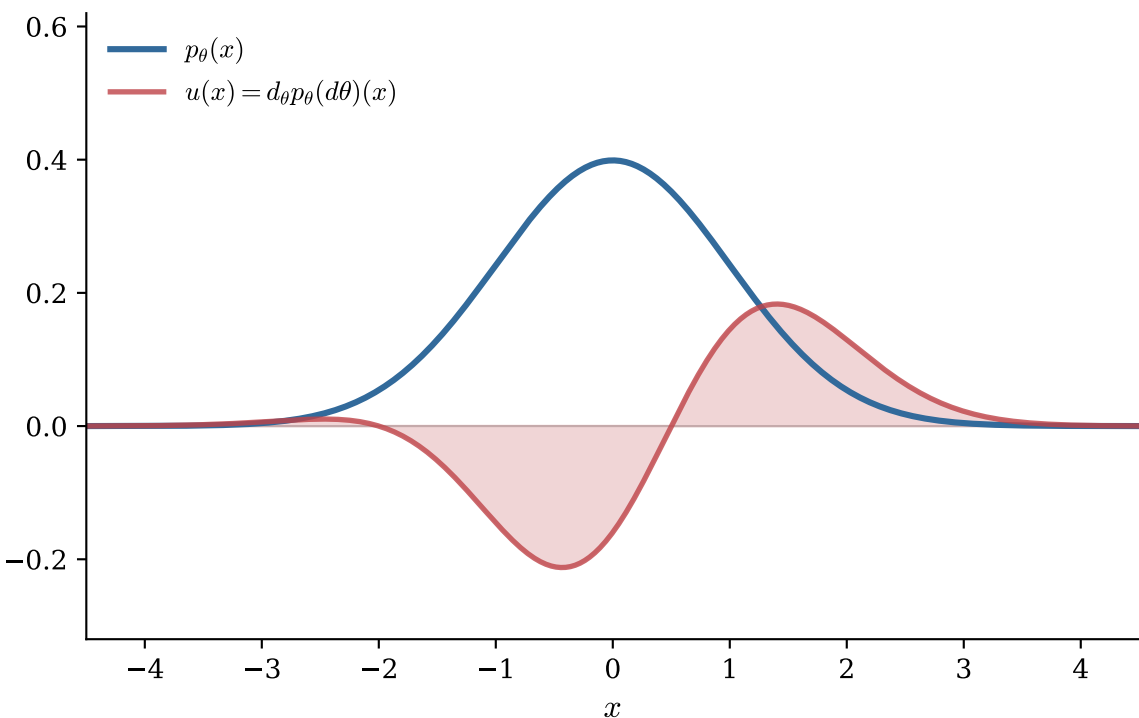


The same tangent direction measured in two spaces

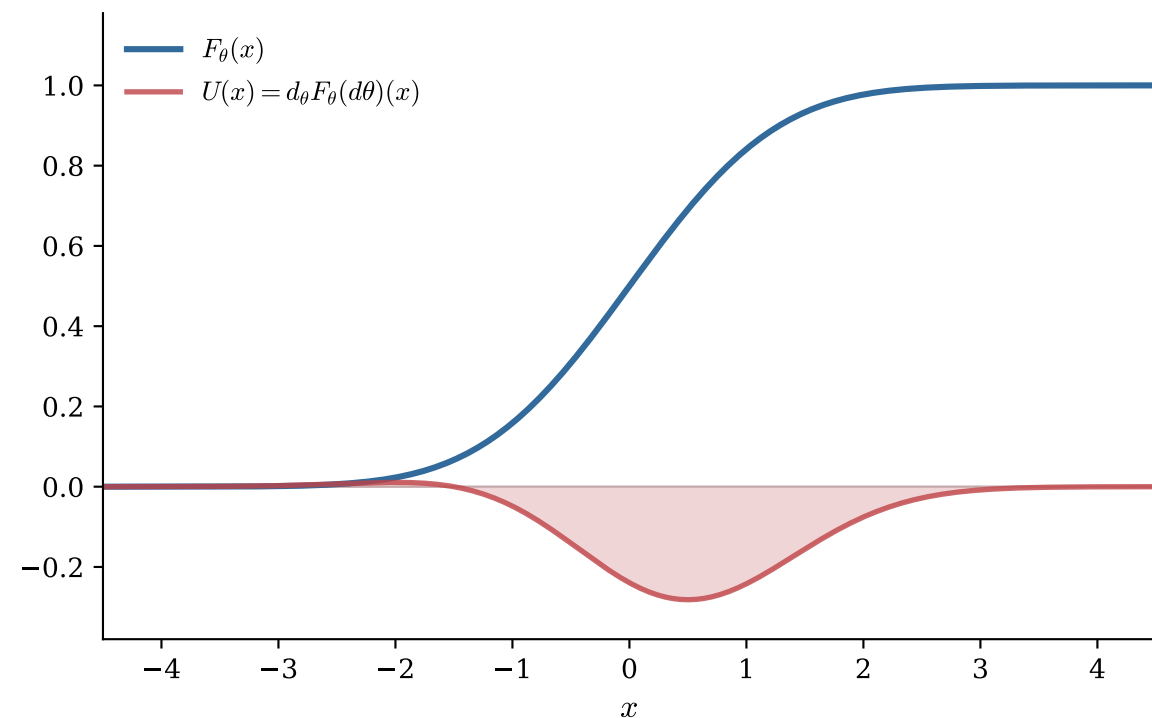
$$p_\theta = \mathcal{N}(\mu, \sigma^2), \quad \theta = (0, 1), \quad d\theta = (0.6, 0.4)$$

$$\|d\theta\|_{g_\theta^{\text{wo}}} = \|d_\theta Q_\theta(d\theta)\|_{L^2(0,1)} = \sqrt{0.52} \simeq 0.7211$$

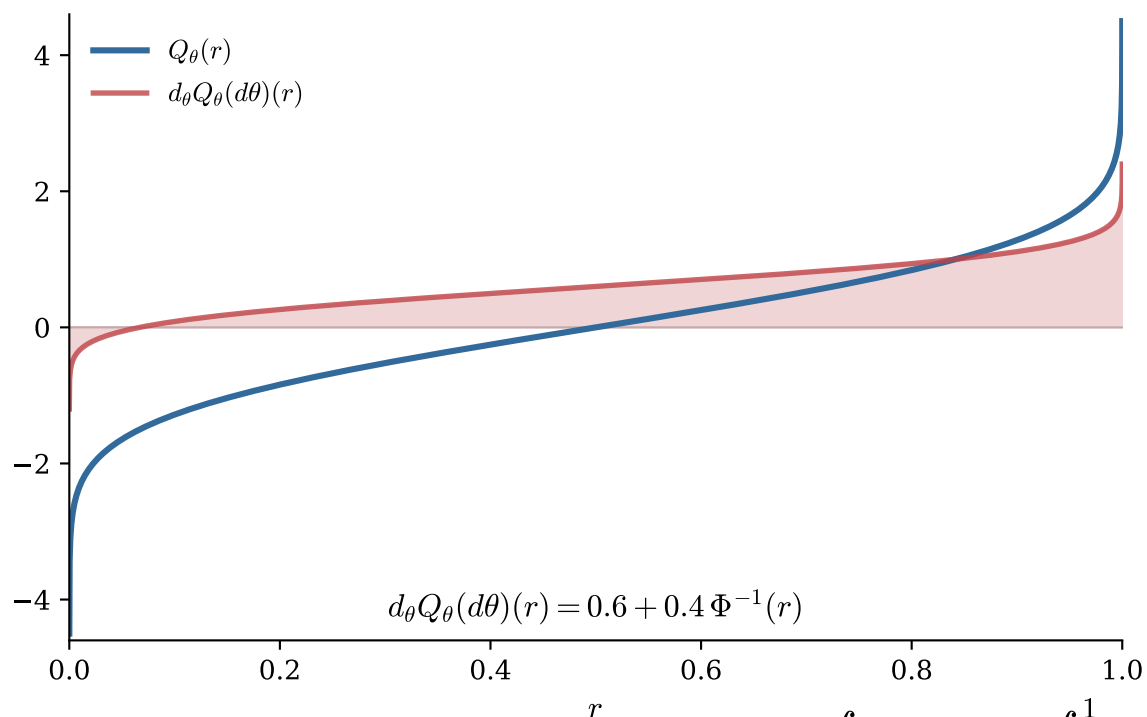
(a) Density and density variation



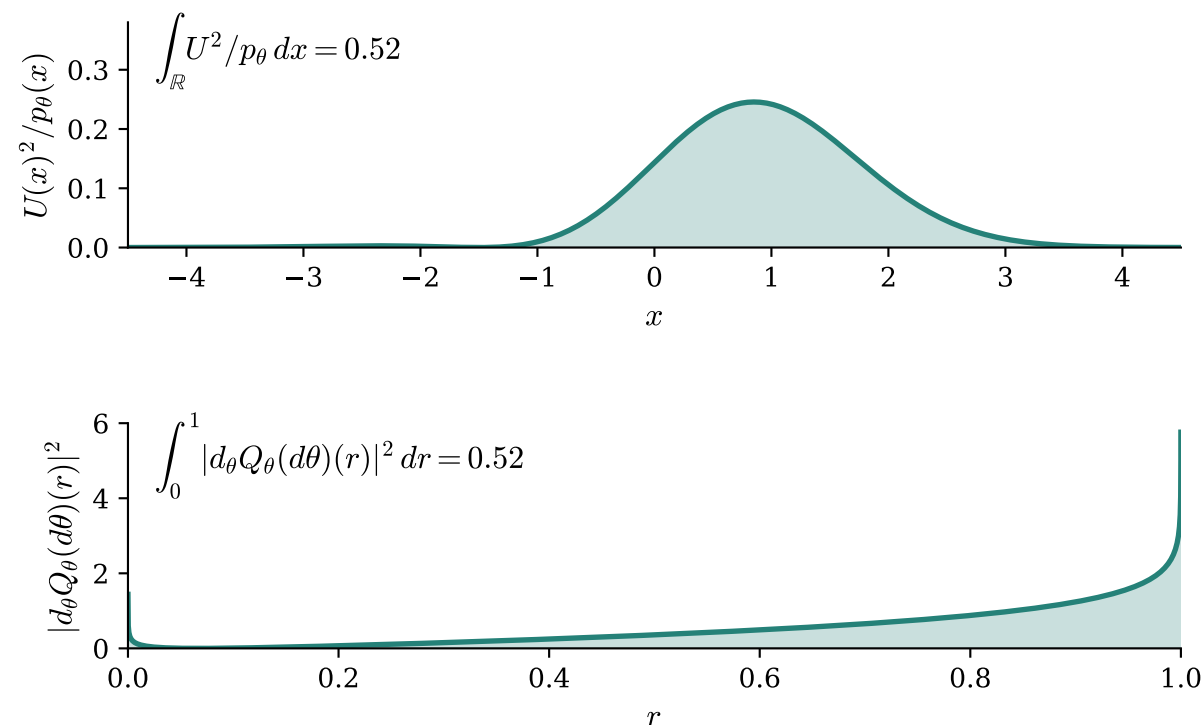
(b) Cumulative function and its variation



(c) Quantile function and its variation



(d) Equal areas: the squared norms



$$\|d\theta\|_{g_\theta^{\text{wo}}}^2 = \int_{\mathbb{R}} \frac{U(x)^2}{p_\theta(x)} dx = \int_0^1 |d_\theta Q_\theta(d\theta)(r)|^2 dr = \|d_\theta Q_\theta(d\theta)\|_{L^2(0,1)}^2$$

Curves are displayed over a finite range; integral values use the full domains.